

Isotopes

Learning outcomes

By the end of this section you should be able to perform calculations involving non-integer relative atomic masses and abundance of isotopes from given data.

	1	2	3	4	5
1	1 H 1.01				
2	3 Li 6.94	4 Be 9.01			
3	11 Na 22.99	12 Mg 24.31			
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94

Atomic number

Element

Relative atomic mass

Figure 1. Layout of element data in the periodic table.

 More information for figure 1

The periodic table in [section 7 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) of the DP chemistry data booklet has a box for each element. At the bottom of each box is a value for the relative atomic mass (symbol A_r). Because atoms have such small masses (in the magnitude of 10^{-27} kg), a relative scale is used. Using these relative values, the masses of atoms can be compared. But where do these values come from? And why are these values decimals, when the numbers of protons and neutrons in an atom are whole numbers?

International Mindedness

The IUPAC definition of relative atomic mass can be found [here](https://goldbook.iupac.org/terms/view/RO5258)  (<https://goldbook.iupac.org/terms/view/RO5258>). IUPAC (the International Union of Pure and Applied Chemistry) is an international organisation that specialises in chemical nomenclature (naming) of compounds as well as naming newly discovered elements. You will come across this naming convention in the naming

of organic compounds in [subtopic S3.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>).

Isotopes are atoms of the same element that have different numbers of neutrons. In other words, they have the same number of protons but a different number of neutrons. Isotopes can be represented by their nuclear symbol, or the name of the isotope followed by its mass number. For example, the isotope of carbon with a mass number of 12 can be represented as $^{12}_6\text{C}$, ^{12}C , or as carbon-12.

Consider the three isotopes of hydrogen (**Figure 2**). Each isotope has one proton in its nucleus as it is the atomic number that gives the atom its identity. They also have the same number of electrons, which is one. What differs, however, is the number of neutrons each isotope has in its nucleus. As you can see from the table, the isotopes of hydrogen contain zero, one and two neutrons respectively.

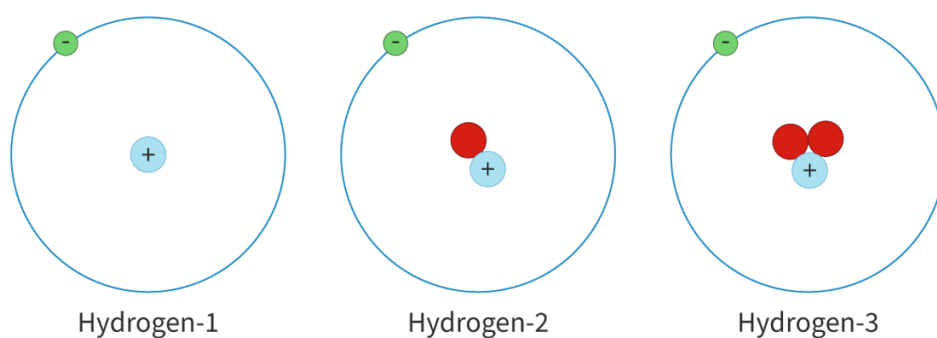


Figure 2. The three isotopes of hydrogen: hydrogen-1 (protium), hydrogen-2 (deuterium) and hydrogen-3 (tritium).

📖 More information for figure 2

Table 1 below shows the numbers of protons, neutrons and electrons in each of the isotopes.

Table 1. Numbers of protons, neutrons and electrons in the isotopes of hydrogen.

Isotope	Number of protons	Number of neutrons	Number of electrons
^1_1H	1	0	1
^2_1H	1	1	1
^3_1H	1	2	1

As you can see from the table, the isotopes of hydrogen contain zero, one and two neutrons respectively. This difference in the number of neutrons has an effect on the physical properties of the isotope. Isotopes of the same element behave

similarly in chemical reactions. The differences in physical properties are caused by the different masses of the isotopes. **Table 2** shows the differences in boiling point, melting point and density for the three hydrogen isotopes.

Table 2. Differences in boiling point, melting point and density for the three hydrogen isotopes

Isotope	Boiling point (K)	Melting point (K)	Density (g cm ⁻³)
${}^1_1\text{H}$	20.4	14.0	0.09
${}^2_1\text{H}$	23.7	18.7	0.18
${}^3_1\text{H}$	25.0	20.6	0.27

What do you notice about the trends in the physical properties of the isotopes of hydrogen?

You can see from **Table 2**, that as the mass number of an isotope increases, so does the melting point, boiling point and density. This is a result of the increasing mass of isotopes as the number of neutrons in the nucleus increases.

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate the skills and recognise the benefits of working collaboratively.
- Demonstrate engagement with issues of global significance.

Certain isotopes are radioactive and are released during nuclear accidents. Two such examples are the nuclear meltdown in Fukushima [🔗 \(https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/fukushima-daiichi-accident.aspx\)](https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/fukushima-daiichi-accident.aspx), Japan in 2011 and the 1986 meltdown in Chernobyl [🔗 \(https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/chernobyl-accident.aspx\)](https://world-nuclear.org/information-library/safety-and-security/safety-of-plants/chernobyl-accident.aspx). In both incidents, large amounts of radioactive material were released into the environment. Despite these risks, nuclear power is still used as an energy source by many different countries.

One possible CAS experience is to raise awareness of the use of nuclear power in the form of a group presentation or poster. You could include a brief overview of its uses as well as including its advantages and disadvantages.

Calculating relative atomic mass, A_r

The relative atomic mass (symbol A_r) of an atom can be calculated from the percent abundance and the masses of the isotopes of that atom. Note that relative atomic mass is covered in more detail in section S1.4.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/relative-atomic-mass-and-relative-formula-mass-id-44261/>). Here we will only learn how to calculate the relative atomic mass of an atom using the mass numbers of the isotopes and their percent abundances. Percent abundance is the percentage of a specific isotope found in a naturally occurring sample of the element. **Table 3** shows the percent abundances of the three naturally occurring isotopes of magnesium (symbol Mg). It should be noted the abundances of the isotopes are given as percentages and therefore add up to 100%.

Look up the relative atomic mass of magnesium in the periodic table. What is the relationship between the mass of the isotope and its percent abundance?

From **Table 3** you can see that most of the naturally occurring sample of magnesium consists of the isotope magnesium-24, therefore we would expect the relative atomic mass of magnesium to be closer to 24 than to 26.

Table 3. Abundance of magnesium isotopes.

Isotope	Abundance (%)
Magnesium-24	78.99
Magnesium-25	10.00
Magnesium-26	11.01

To calculate the relative atomic mass, the mass number of each isotope is multiplied by its percent abundance. These values are then added together and then divided by 100 to give the relative atomic mass of a single atom. Using the data in **Table 3**:

$$\begin{aligned}\text{Relative atomic mass of magnesium} &= \frac{(24 \times 78.99) + (25 \times 10.00) + (26 \times 11.01)}{100} \\ &= 24.32\end{aligned}$$

As you can see from this calculation, the relative atomic mass of magnesium is closer to 24 because the isotope with a mass of 24 has the highest percent abundance.

Worked example 1

Use the data in the table to calculate the relative atomic mass of titanium correct to two decimal places.

Isotope	Abundance (%)
Titanium-46	8.25
Titanium-47	7.44
Titanium-48	73.72
Titanium-49	5.41
Titanium-50	5.18

$$A_r = \frac{(46 \times 8.25) + (47 \times 7.44) + (48 \times 73.72) + (49 \times 5.41) + (50 \times 5.18)}{100}$$

$$= 47.92$$

The relative atomic mass of titanium is closer to 48 because the isotope with a mass of 48 has the highest percent abundance.

The percent abundances of the isotopes can be calculated if the relative atomic mass of the atom is known together with the mass numbers of the isotopes.

Worked example 2

Bromine exists as two isotopes: bromine-79 and bromine-81. The relative atomic mass of bromine is 79.90. Use this information to calculate the percent abundances of the two isotopes.

$$79.90 = \frac{(81x + 79(100 - x))}{100}$$

$$7990 = 81x + 79(100 - x)$$

$$7990 = 81x + 7900 - 79x$$

$$7990 - 7900 = 81x - 79x$$

$$90 = 2x$$

$$x = 45$$

$$\begin{aligned} \text{Percent abundance of bromine-79} &= 100 - x \\ &= 100 - 45 \\ &= 55\% \end{aligned}$$

$$\begin{aligned} \text{Percent abundance of bromine-81} &= x \\ &= 45\% \end{aligned}$$

Study skills

When calculating the relative atomic mass of an element, the answer is commonly given to two decimal places. However, make sure you read the question carefully to make sure that you give the answer to the number of decimal places or significant figures stated.

Always repeat the calculation to make sure you did not make a mistake with brackets or digits.

In this activity you will perform calculations involving the relative abundance of isotopes.

Activity

- **Learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Reflecting on the credibility of results
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual

1. Calculate the relative atomic mass of gallium to two decimal places. Before you do the calculation, make a prediction based on the mass and the percent abundance of the isotopes.

Isotope	Abundance
^{69}Ga	60.10
^{71}Ga	39.90

Silver has two naturally occurring isotopes; ^{107}Ag and ^{109}Ag . The relative atomic mass of silver is 107.87. Determine the percent abundance of each isotope.

Once you have completed these two examples, here is a [link](https://teachchemistry.org/classroom-resources/isotopes-calculating-average-atomic-mass-simulation)  (<https://teachchemistry.org/classroom-resources/isotopes-calculating-average-atomic-mass-simulation>) to a simulation for further practice.

$$1. \text{ Relative atomic mass of gallium} = \frac{(69 \times 60.10) + (71 \times 39.90)}{100} = 69.80$$

$$\begin{aligned}
 2. \quad 107.87 &= \frac{109x + 107(100 - x)}{100} \\
 10787 &= 109x + 10700 - 107x \\
 10787 - 10700 &= 109x - 107x \\
 87 &= 2x \\
 x &= 43.5
 \end{aligned}$$

$$\begin{aligned}\text{Percent abundance of } ^{109}\text{Ag} &= x \\ &= 43.5\% \\ \text{Percent abundance of } ^{107}\text{Ag} &= 100 - x \\ &= 56.5\%\end{aligned}$$